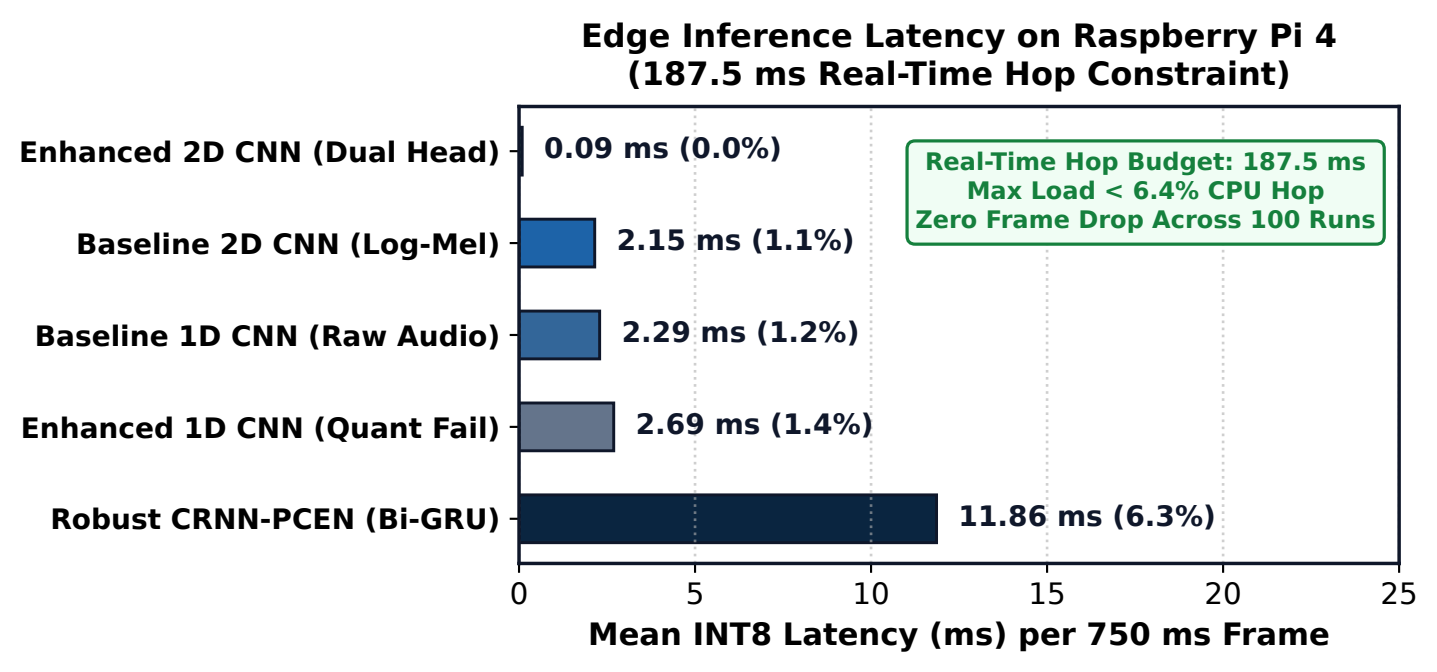
Edge Inference Latency on Raspberry Pi 4
(187.5 ms Real-Time Hop Constraint)
Enhanced 2D CNN (Dual Head)
0.09 ms (0.0%)
Baseline 2D CNN (Log-Mel)
2.15 ms (1.1%)
Baseline 1D CNN (Raw Audio)
2.29 ms (1.2%)
Enhanced 1D CNN (Quant Fail)
2.69 ms (1.4%)
Robust CRNN-PCEN (Bi-GRU)
11.86 ms (6.3%)
Real-Time Hop Budget: 187.5 ms
Max Load < 6.4% CPU Hop
Zero Frame Drop Across 100 Runs
0
5
10
15
20
25
Mean INT8 Latency (ms) per 750 ms Frame